A Theory of Economic Slack

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Draft version: July 2026

APPENDIX B.

Elasticities

In this appendix, we review the concept of elasticity, which we use repeatedly throughout the book, and we provide a set of results that are useful to handle elasticities. Throughout we consider a function $f(x)$ that is strictly positive and differentiable for $x > 0$.

B.1. Definition of elasticities

The elasticity of $f(x)$ with respect to x, denoted ϵ^f_x, is defined as

$$\epsilon^f_x = \frac{x}{f(x)} \cdot f'(x), \tag{B.1}$$

where $f'(x)$ is the standard derivative of f. All properties of elasticities follow from this definition.

B.2. Leibniz notation for differentials

To work with elasticities, it is convenient to introduce Leibniz's notation for differentials. In this notation, dx denotes an infinitesimal change in a variable x, and $df(x)$ denotes an infinitesimal change in a function $f(x)$. The differential $df(x)$ is related to the differential dx and the derivative $f'(x)$ of the function f by

$$df(x) = f'(x)\, dx.$$

From this definition, we see that the differential operator d obeys the same algebraic rules as derivatives.

The Leibniz notation is particularly convenient to use with the logarithm. Given that the derivative of $\ln(x)$ is $1/x$:

$$d\ln(x) = \frac{dx}{x}. \tag{B.2}$$

Hence, while dx measures the absolute change in x, $d\ln(x)$ measures a relative change in x.

Applying the logarithm to the function f, we obtain

$$d\ln f(x) = \frac{df(x)}{f(x)}. \tag{B.3}$$

So while $df(x)$ measures the absolute change in $f(x)$, $d\ln f(x)$ measures a relative change in $f(x)$.

These equations show that the log-differentials $d\ln(x)$ and $d\ln f(x)$ are simply a compact way of writing the relative change in the variable x or the function $f(x)$.

B.3. Interpretation of elasticities

What does an elasticity mean? To answer this, we rewrite the definition (B.1) using the Leibniz notation $f'(x) = df/dx$:

$$\epsilon_x^f = \frac{x}{f(x)} \cdot \frac{df(x)}{dx} = \frac{df(x)/f(x)}{dx/x}. \tag{B.4}$$

We see from this equation that the elasticity is the ratio of the relative change in f to the relative change in x.

This gives a simple interpretation. Consider a 1% change in x, which means its relative change is $dx/x = 0.01$. The resulting relative change in $f(x)$ is:

$$\frac{df(x)}{f(x)} = \epsilon_x^f \times \frac{dx}{x} = \epsilon_x^f \times 0.01.$$

Thus, a 1% change in x leads to an ϵ_x^f percentage change in $f(x)$: the elasticity measures the percentage response of a function to a 1% change in its input.

B.4. Connection between elasticities and log-differentials

Elasticities are so convenient because of their connection to natural logarithms. This connection comes from a basic result in calculus: the derivative of $\ln(x)$ is $1/x$. In Leibniz's notation, this means $d\ln(x) = dx/x$, as we saw in equation (B.2). Using this insight and expression (B.4), we can express the elasticity as the ratio of two log-differentials. This is not a new definition, but a convenient reformulation of (B.1):

$$\epsilon_x^f = \frac{d\ln f(x)}{d\ln(x)}. \tag{B.5}$$

This logarithmic form often helps compute the elasticity of complex functions.

Reshuffling (B.5), we also see that the elasticity relates the log-differential of the function to the log-differential of the argument:

$$d\ln f(x) = \epsilon_x^f \, d\ln(x).$$

B.5. Useful results on elasticities

We now introduce results that we use in the book to compute elasticities. Throughout, the functions $g(x)$ and $h(x)$ are assumed to be strictly positive and differentiable for $x > 0$.

RESULT B.1. *For any real a, the elasticity of the function $f(x) = x^a$ is $\epsilon_x^f = a$.*

PROOF. Since $\ln(f(x)) = a\ln(x)$, differentiating gives $d\ln(f(x)) = a\,d\ln(x)$. Dividing by $d\ln(x)$, we get $d\ln(f(x))/d\ln(x) = a$. We obtain the result by applying (B.5). □

RESULT B.2. *For any $a > 0$, the elasticity of the function $f(x) = a \cdot g(x)$ is $\epsilon_x^f = \epsilon_x^g$.*

PROOF. Since $\ln(f(x)) = \ln(a) + \ln(g(x))$, and $\ln(a)$ is just a constant, then differentiating gives $d\ln(f(x)) = d\ln(g(x))$, so $d\ln(f(x))/d\ln(x) = d\ln(g(x))/d\ln(x)$. Once again, we obtain the result by applying (B.5). □

RESULT B.3. *The elasticity of the function $f(x) = g(x) \cdot h(x)$ is $\epsilon_x^f = \epsilon_x^g + \epsilon_x^h$.*

PROOF. Since $\ln(f(x)) = \ln(g(x)) + \ln(h(x))$, then $d\ln(f(x)) = d\ln(g(x)) + d\ln(h(x))$. We reach the result after dividing by $d\ln(x)$ and using (B.5). □

RESULT B.4. *The elasticity of the function $f(x) = g(x)/h(x)$ is $\epsilon_x^f = \epsilon_x^g - \epsilon_x^h$.*

PROOF. Since $\ln(f(x)) = \ln(g(x)) - \ln(h(x))$, then $d\ln(f(x)) = d\ln(g(x)) - d\ln(h(x))$. Again, we obtain the result after dividing by $d\ln(x)$ and using (B.5). □

RESULT B.5. *The elasticity of the function $f(x) = g(x) + h(x)$ is*

$$\epsilon_x^f = \frac{g(x)}{g(x) + h(x)} \cdot \epsilon_x^g + \frac{h(x)}{g(x) + h(x)} \cdot \epsilon_x^h.$$

PROOF. For sums of functions, the logarithmic definition of elasticities is not helpful, so we return to the original definition, given by (B.1). Since $f'(x) = g'(x) + h'(x)$, we get

$$\begin{aligned}
\epsilon_x^f &= \frac{x}{f(x)} \cdot f'(x) \\
&= \frac{x}{g(x)+h(x)} \left[g'(x) + h'(x)\right] \\
&= \frac{g(x)}{g(x)+h(x)} \left[\frac{x}{g(x)} \cdot g'(x)\right] + \frac{h(x)}{g(x)+h(x)} \left[\frac{x}{h(x)} \cdot h'(x)\right].
\end{aligned}$$

We obtain the result from here, as the definition of elasticities also gives $\epsilon_x^g = xg'(x)/g(x)$ and $\epsilon_x^h = xh'(x)/h(x)$. □

RESULT B.6. *The elasticity of the function $f(x) = g(x) - h(x)$, with $h(x) < g(x)$, is*

$$\epsilon_x^f = \frac{g(x)}{g(x)-h(x)} \cdot \epsilon_x^g - \frac{h(x)}{g(x)-h(x)} \cdot \epsilon_x^h.$$

PROOF. Here again, we apply the elasticity definition (B.1). Since $f'(x) = g'(x) - h'(x)$, we get

$$\epsilon_x^f = \frac{x}{g(x)-h(x)} \left[g'(x) - h'(x)\right] = \frac{g(x)}{g(x)-h(x)} \left[\frac{x}{g(x)} \cdot g'(x)\right] - \frac{h(x)}{g(x)-h(x)} \left[\frac{x}{h(x)} \cdot h'(x)\right],$$

which again yields the result. □

RESULT B.7. *The elasticity of the function $f(x) = g(h(x))$ is*

$$\epsilon_x^f = \epsilon_x^h \cdot \epsilon_h^g,$$

where ϵ_h^g is the elasticity of g evaluated at $h(x)$.

PROOF. By the chain rule, $f'(x) = h'(x) \cdot g'(h(x))$. Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{g(h(x))} \cdot h'(x) \cdot g'(h(x)) = \left[\frac{x}{h(x)} \cdot h'(x)\right]\left[\frac{h(x)}{g(h(x))} \cdot g'(h(x))\right].$$

The result follows from here as $\epsilon_x^h = xh'(x)/h(x)$ and $\epsilon_h^g = h(x)g'(h(x))/g(h(x))$. □

RESULT B.8. *Assume that the function $g(y)$ is invertible with differentiable inverse. The elasticity of the function $f(x) = g^{-1}(x)$ is*

$$\epsilon_x^f = \frac{1}{\epsilon_y^g},$$

where ϵ_y^g is the elasticity of g evaluated at $y = g^{-1}(x)$.

PROOF. By the inverse function theorem, $f'(x) = 1/g'(g^{-1}(x))$. Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{g^{-1}(x)} \cdot \frac{1}{g'(g^{-1}(x))}.$$

The result follows because $\epsilon_y^g = yg'(y)/g(y)$ with $y = g^{-1}(x)$. □

Finally, we introduce a bivariate function $k(y,z)$ which is strictly positive and differentiable for $y > 0$ and $z > 0$. We also assume that the arguments y and z are themselves functions of x, and the functions $y(x)$ and $z(x)$ are strictly positive and differentiable for $x > 0$.

RESULT B.9. *The elasticity of the function $f(x) = k(y(x), z(x))$ is*

$$\epsilon_x^f = \epsilon_y^k \cdot \epsilon_x^y + \epsilon_z^k \cdot \epsilon_x^z,$$

where ϵ_y^k and ϵ_z^k are partial elasticities of the function k:

$$\epsilon_y^k = \frac{y}{k(y,z)} \cdot \frac{\partial k}{\partial y} \quad \text{and} \quad \epsilon_z^k = \frac{z}{k(y,z)} \cdot \frac{\partial k}{\partial z}.$$

PROOF. By the multivariate chain rule,

$$f'(x) = \frac{\partial k}{\partial y} \cdot y'(x) + \frac{\partial k}{\partial z} \cdot z'(x).$$

Hence, using the elasticity definition (B.1), we get

$$\epsilon_x^f = \frac{x}{k(y,z)} \left[\frac{\partial k}{\partial y} \cdot y'(x) + \frac{\partial k}{\partial z} \cdot z'(x)\right] = \left[\frac{y}{k(y,z)} \cdot \frac{\partial k}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right] + \left[\frac{z}{k(y,z)} \cdot \frac{\partial k}{\partial z}\right]\left[\frac{x}{z} \cdot z'(x)\right].$$

We obtain the result from this equation and the definitions of the standard and partial elasticities. □

B.6. Implicit differentiation with elasticities

Implicit differentiation is very helpful to compute derivatives of functions that are defined implicitly. It turns out that we can also use the same technique to compute elasticities of functions defined implicitly.

RESULT B.10. *Assume that the function $y(x) > 0$ is defined implicitly by the equation $f(x, y) =$*

$g(x, y)$, where the bivariate functions $f(x, y)$ and $g(x, y)$ are strictly positive and differentiable for $x > 0$ and $y > 0$. Taking elasticities on both sides of the equation, the elasticity of y with respect to x satisfies

$$\epsilon_x^f + \epsilon_y^f \cdot \epsilon_x^y = \epsilon_x^g + \epsilon_y^g \cdot \epsilon_x^y,$$

where ϵ_x^f, ϵ_y^f, ϵ_x^g, ϵ_y^g, are partial elasticities defined by

$$\epsilon_x^f = \frac{x}{f(x, y)} \cdot \frac{\partial f}{\partial x}, \qquad \epsilon_y^f = \frac{y}{f(x, y)} \cdot \frac{\partial f}{\partial y},$$
$$\epsilon_x^g = \frac{x}{g(x, y)} \cdot \frac{\partial g}{\partial x}, \qquad \epsilon_y^g = \frac{y}{g(x, y)} \cdot \frac{\partial g}{\partial y}.$$

Collecting terms, we can express the elasticity of y as a function of the partial elasticities of the functions f and g:

$$\epsilon_x^y = \frac{\epsilon_x^g - \epsilon_x^f}{\epsilon_y^f - \epsilon_y^g}.$$

PROOF. The implicit differentiation of the equation $f(x, y) = g(x, y)$ gives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot y'(x) = \frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \cdot y'(x).$$

We then divide the left-hand side by $f(x, y)$ and the right-hand side by $g(x, y)$, which we can do because $f(x, y) = g(x, y)$, and we multiply both sides by x. We get

$$\frac{x}{f(x, y)} \cdot \frac{\partial f}{\partial x} + \left[\frac{y}{f(x, y)} \cdot \frac{\partial f}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right] = \frac{x}{g(x, y)} \cdot \frac{\partial g}{\partial x} + \left[\frac{y}{g(x, y)} \cdot \frac{\partial g}{\partial y}\right]\left[\frac{x}{y} \cdot y'(x)\right].$$

We obtain the result from this equation and the definitions of the standard and partial elasticities. □